Intelligent software technologies

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During the last four decades the field of artificial intelligence has made impressive progress. It is true that not all the promises generated during the enthusiasm of the early years have yet been delivered. But the fact remains that today's systems do provide impressive levels of machine intelligence. There are many examples of intelligent software systems which can match human capabilities. However, we must not fall into the trap of wanting computer systems to emulate fully human performance, for therein lie many subtle philosophical, scientific and engineering pitfalls. This paper provides a personal account of the progress of AI and offers an optimistic viewpoint.

1. Introduction

We are poised at a threshold — a threshold that clearly delineates two epochs in human development. All history that preceded us will, in the future, characterise man as the tool user. At some point in the near future mankind will, for the first time, enjoy a cognitive partnership, a symbiotic relationship with its own creation — the intelligent computer. Fanciful ideas maybe, but all the indications are that the past 50 years or so of research into artificial intelligence are about to bear fruit; and the product of this long and difficult development period will have a profound impact on our lives and future development as a species.

So why do we take this upbeat account of artificial intelligence (AI) research? Is it because there have been spectacular breakthroughs in AI? No. Could it be that there is a universal demand for the products and intuitions of this branch of system engineering and computer science? Again the answer must still be negative. But, and this is the telling argument, what the objective observer should recognise is that the field is making significant incremental advances on a broad front. These advances include:

- the ability to recognise human speech and the meanings contained therein,
- the ability of computer systems to infer information from visual scenes,
- the power of expert systems and knowledge-based systems to undertake very complex diagnostic and analytic tasks,
- the manner in which computer programmes now intrude into human games such as chess and cards,
- the way simple code helps us plan our daily lives and assists us both with routine tasks and in a multitude of other ways where we are now relying on (intelligent) computer systems.

Admittedly, many people will find this upbeat account unacceptable. However, if we go back only a relatively short period of time, we discover that the commonly held view was that people could never travel faster than the speed of a horse; and it was only a few centuries ago that the geocentric universe held sway.

In this paper we want to set out our arguments why we believe an information processing revolution is at hand. We adopt a frankly tendentious stance, while appreciating that there are many eminent scientists and philosophers who are dismissive of the field. We accept the sincerity of these deep mathematical and philosophical arguments but counter-attack with the engineering viewpoint which has been our life-long perspective.

Most of this paper will address the technical and philosophical boundaries that define AI research. We outline both the important (incremental) advances and the significant hurdles yet to be surmounted. We draw from both mainstream AI work and, where necessary to illustrate a point, from the writings of philosophers who have addressed similar issues. Section 2 reviews the overriding need for intelligent systems in our information rich society. Section 3 gives background ideas drawn from the supporting philosophical tradition. Section 4 provides a brief and deliberately incomplete review of the current state of intelligent systems and section 5 gives the other side of the coin by describing the major hurdles yet to be mastered. The paper concludes with a short discussion and an optimistic view of future progress.

2. The need for intelligent software systems

So why is it that in the closing years of the 20th century the need for intelligent software systems is becoming urgent. The simple answer is, of course, that the world is in the midst of an information revolution. Increasingly all data and information is being captured, transported and processed digitally. The fruits of this digital revolution, which provide instant access to the vast global data warehouse, has brought with it gross data overload. Our human cognitive capabilities, such as memory and comprehension, have not developed (and we would conjecture will not develop) to keep pace with our own invention. What this means is that we now need assistance from intelligent software systems in order to understand the world we are in the process of creating. These systems are intelligent in the sense that they have the capability to conduct information processing tasks which if performed by humans would require a high degree of skill and not inconsiderable intelligence.

A great deal has been written on the need for intelligent systems [1, 2]. For our purposes we can recall a few of these:

- to assist with routine but important tasks,
- to undertake complex tasks rapidly,
- to act as gofers on our behalf,
- to explore large cognitive search spaces to find elusive solutions,
- to deal with the data-rich environment which we have created.

What underpins the systems-engineering approach to all these intelligent software systems to a greater or lesser extent is the desire to capture some aspect of human intellectual capability. Such a desire may be formally stated (the human/computer interface community) or be a trivial anthropomorphic device (smiling face). The reason for this homocentric approach is the recognition that we humans are most comfortable when interacting with other humans. However, it has to be admitted that in certain circumstances, usually of highly personal nature, some people prefer conversing with machines. Therefore we are not pretending that intelligent software-systems technology offers solutions to all human information processing needs — only a large proportion of them.

As the goal of human-oriented computing is approached, the expectation of combining digital speed and precision with human levels of intuition grows.

3. Philosophical background

Research in artificial intelligence has been founded on the existence theorem which says that if a task can be done by a natural physical organ, i.e. the brain, then it should be possible to devise a man-made device to carry out that task. An additional hypothesis that is usually made is that the best candidate for the man-made device is the digital computer. Even if not explicitly stated, some such belief provides the underlying motivation to all workers in AI. However, as stated above, the theorem begs many questions and lays itself open to challenges on several fronts. Penrose's challenge [3], in simple terms, is just that while the brain is no doubt a purely physical organ, it is governed by physical laws that are little understood at present; that these laws must be related to the finer points of quantum theory and as such AI research is in no position to realise all its ambitions. Dreyfus's challenge [4] is a little different, and appears to us to question that part of the theorem that posits all human intelligence to a single organ, namely, the brain. If this reading is correct, then our position is not too far removed from his; but we argue as engineers and not as AI theoreticians; and Dreyfus does not have any quarrels with an engineering approach to intelligent software systems. We will have more to say on this matter later. As to the additional hypothesis, Dreyfus seems to say that the von Neuman architecture (serial computing) must be an important limiting factor on the conventional AI research programme. Penrose does not consider that to be an important issue, since artificial neural networks (parallel computing) as used by AI workers can always be simulated on a conventional digital computer.

One of the most important questions that the existence theorem raises is indeed concerned with the organ in question, the brain. Each individual's organ is constructed bio-chemically based on the blueprint provided by the owner's genetic material. That blueprint has been subjected to Darwinian evolution and so it is not surprising that while all animals possess a brain, the human animal displays uniquely more powerful intellectual abilities. However, the fact that human intellectual competence varies so widely from one individual to another cannot be explained away easily as purely genetic, although many would have us believe this to be so. Furthermore, the continuing progress of human knowledge since man first evolved, cannot be explained in terms of a continuing evolution of the human brain.

Humans have been around for about 100 000 years or 4000 generations. This is too short a period for the brain to have evolved very much. The 'out of Africa' theory is now well accepted; that the human race evolved in one place — Africa — and then migrated to other parts of the world. There were other humanoid life-forms such as the Neanderthal man, but these did not survive. We assume the human differed from other humanoids because he had a more elaborate language. A distant predecessor of *homo-sapiens*, the *homo erectus* was so called because he was a biped, with two forelimbs available for purposes other than locomotion. He was followed by *homo habilis*, so called because he had dexterous hands which he used for making tools. It is clear that what characterised the *homo sapiens* is that he evolved another set of equally dexterous muscles — the muscles of the mouth; and that is what enabled speech. Human intelligence is attributed to the much larger brain of the *homo sapiens* compared to other homonids. However, it is not this final adult size that is important but the fact that it comes about through the process called neoteny. This process results in the size of the infant brain being only 25% of its final adult size. That is to say the brain grows fourfold after birth and, most importantly, develops in the presence of its environment.

The above facts should lead us to the conjecture that the human adult brain of each individual consists of a complex set of predispositions to act (a wired response, if you will) that is conditioned by that individual's environment, or more precisely from his sensitivities to that environment at every stage of his early development. The many permutations of the environment that we individually encounter, and our constantly changing emotional state, and thus the sensitivity to that environment, is what ultimately produces such a wide spectrum of intellectual competence among us humans. As someone put it in a semi-humourous way, if we clone a number of individuals from Einstein's genetic material, we may not get a number of physics geniuses but only a whole lot of patent office clerks. This would suggest that all brains are substantially the same and that the variation in human competence is predominantly down to environmental factors. That ought to be good news for the supporters of artificial intelligence research.

One of the principal results of the developmental process in an individual is the acquisition of a wide variety of elementary skills. These are largely in the form of sub-conscious habits and often imprinted as what can be described as 'muscle memory'. Therefore, a significant number of skills, habits or muscle memory acquired during the developmental process refer to the muscles of the mouth, i.e. speech. It is noteworthy that people with amnesia may forget many things about themselves, but not the basic skills like walking and talking their mother tongue, indicating that these are a wired or response-muscle memory. It is not too far fetched to say that the human mouth is as important as the human brain for our superior intelligence. Could it be that we should properly call ourselves *homo loquax*, the garrulous man?

Language is the result of our ability to speak. It is primarily a time-invariant regularity for associating sound to things, events, etc. It is this regularity of sounds, or language that gave us the ability to retain the intermediate results of thought and to recall them for further thought — a kind of short-term memory. Gradually it formed our growing and evolving knowledge. The evolution of knowledge and passing it reliably from one generation to the next was greatly increased when humans developed written language somewhere around 5000 years, or 200 generations ago.

We can safely say that while the human brain itself has not evolved over the 4000 generations the knowledge that human society possesses has steadily (and recently rapidly) increased. Indeed most of human evolution over that period has occurred outside the human body — in human knowledge. This has been best described by Sir Karl Popper. Popper is known as an important philosopher of science, but it is his ideas about World 3 [5] that have a very significant relevance for research workers in AI.

According to Popper, World 1 is 'the world of physical objects, or the world of physical states'. World 2 is 'the world of states of consciousness, or the world of mental states, or the world of dispositions to act'. Now most AI workers would contend that the field investigates, or is inspired by, the phenomena of this World 2. However, Popper maintains that we also have a World 3 which is the world of objective content of scientific thought, poetic thought and works of art. Other contents of World 3 are theoretical systems, problems, problem situations, critical arguments, the contents of journals or books and so on. He says: "Traditional epistemology has studied knowledge or thought in a subjective sense ... While knowledge in the sense of .. 'I know' belongs to World 2, the world of subjective thoughts; scientific knowledge belongs to World 3, the world of objective theories, objective problems and objective arguments."

Popper gives various theses regarding World 3 that are all relevant here:

- traditional epistemology based on World 2 is irrelevant to the study of scientific knowledge,
- the study of a largely autonomous World 3 of objective knowledge is of decisive importance for epistemology,
- an objective epistemology that studies World 3 can help to throw an immense amount of light upon World 2 ... , but the converse is not true,

- World 3 is a natural product of the human comparable to a spider's web (Dawkins [6] calls this the extended phenotype).

World 3 is largely autonomous; it is through this interaction between ourselves and World 3 that objective knowledge grows and there is close analogy between the growth (or the evolution) of knowledge and biological evolution [7].

This shows World 3 to be quite different from Plato's ideal knowledge. It is man-made, composed of conjectures, subject to falsification; it is full of imperfections, antinomies and paradoxes. As scientists and engineers we need to treat it with humility and there can never be room for arrogance and certainty. All human intellectual evolution has thus taken place in this Popperian World 3. Neoteny ensures that each child grows up with a different (and hopefully richer and more accurate) World 3 environment. The knowledge an individual forms is then determined by a variety of factors like the mood he is in while he gets introduced to this barrage of World 3 and the sequence in which this exposure takes place. Then, as knowledge evolves, each generation's view of the world changes.

One of the important constituents of this World 3 are models that:

- describe how the physical world around us works,
- prescribe how we should, as social animals, behave; these prescriptive models are pure conjectures and the result of our fertile imagination.

Let us consider the descriptive models first.

Descriptive models — since man began to talk he must have conjectured how the regularities of the physical world came about. These were little different from what we take today to be scientific theories. In time, once the foundations of mathematics were laid, we began creating mathematical models. It took a long time indeed for us to begin to put down the foundations of the scientific method. This method is no more than a few hundred years old and it is the subject of Popper's main enquiry; the part of his work that most lay people are aware of. For him, for any theory (i.e. a descriptive model that is supposed to be about the real world) to be scientific it must be falsifiable. Furthermore, Popper says that there are many descriptive theories that are immunised against falsification. Our World 3 contains many descriptive models that do not measure up to such scientific rigour.

Prescriptive models — we like to think that our prescriptions for day-to-day actions and future behaviour are based upon a knowledge of how the world is; that is to say they are derived from sound descriptive models. Ask anyone why one should behave in some accepted fashion and the tendency will be to reply that because the world is like thus and so. The sad fact is that there are many prescriptive formulae that are not based on any well-founded descriptive knowledge. Often the description is a *post facto* justification for holding on to a given recipe for behaviour; but that is not to say that prescriptive recipes are purely arbitrary and any recipe will therefore do. They have to work, which in fact means that they have to give us an added survival advantage. Having to be fit for the purpose in an economic way, they are thus subject to selection pressure. Prescriptive models are no different from any other invention, and as such it can often happen that although originally intended for one purpose, an invention is actually discovered to be best suited for an entirely different purpose.

Unfortunately many scientists themselves are not very clear about this. Most of the field of computer science is on close scrutiny dealing with prescriptive models (i.e. inventions) of implementing something. Furthermore, given that so much of our knowledge is concerned with models, such models themselves become for many, matters of interest in their own right. Many scientists cannot tell when their focus of study is the structure of the model itself (particularly when the models are expressed mathematically) and when the main interest is to do with the relationship of the models to reality. Much of orthodoxy, in science as elsewhere, arises from a concern to hang on to particular models irrespective of reality.

Dreyfus is correct in focusing his main criticisms on good old fashioned artificial intelligence (GOFAI). Traditionally, AI has devoted more time to human-like sensory performance, rather than human-like motor performance (robotics being the main exception). However, in a major way, it is precisely our muscular dexterity that marks us out as different from other animals. Dogs for example have better hearing than humans, birds of prey have better eyesight than humans, but no animal has as much muscle control over its hands and mouth muscles as we humans do. The brain is important because it co-ordinates and amplifies the best mouth muscles of any animal with a rather mediocre ear, within the animal kingdom. The computer can generate speech, and it can analyse speech, but the computer is seldom made to hear itself speak; the computer brain has so far not been used to co-ordinate the speech input with speech output.

The result is that, at present, our computers have no bodies, i.e. no muscular movements through which they can respond to sense inputs. In the terms used by one faction of AI practitioners, most computer systems are not grounded. These issues are further discussed towards the end of this paper.

4. Progress to date

Over the past four decades, systems embodying techniques from the field of artificial intelligence have enjoyed spectacular advances. These advances have, of course, been symbiotic with the equally impressive strides made in information technology. However, it has to be admitted that delivered systems do not yet live up to the expectations generated by the earlier enthusiasm.

In the following paragraphs a brief, specially selected, account is given of the current state of the art. These examples are intended to demonstrate the healthy condition of AI as a set of techniques and systems engineering approaches to delivering intelligent software systems.

4.1 Human/computer interface

A new user of today's personal computer (PC) or workstation, would find it difficult to comprehend the vast improvement made in the human/computer interface (HCI). When we began utilising computers the 80 column punched card was king! Those men and women who had to program computers in the early days were equally disadvantaged — machine code, or assembly language, was the only option. Today computers are instructed by natural language (text or speech); blocks of text or graphics can be moved or modified by the click of a mouse; following a few keystrokes the World Wide Web provides rapid access to a vast global database; television quality moving images can be called up to illustrate a point or present complex data, etc. On the programming front the progress has been equally spectacular. Relatively short commands in modern high-level languages evoke complex actions by the computer. The compiler performs many levels of consistency checking that was once the bane of programmers.

This level of 'user friendliness' is now expected and provided on entry-level PCs. However, what is by no means generally appreciated is that many of the accepted aspects of computing are the fruits of research conducted in artificial intelligence laboratories back in the 1950s and 1960s. For example, Lisp, the oldest high-level language, is still the language of choice for much current AI research and delivered systems; and now, of course, we are seeing the ready acceptance of agents technology [8] to further adapt the computer to a more human-like entity.

As we noted in section 3, language is crucial to human intellectual development. Therefore it is not surprising that natural language processing (NLP) has been a major pre-occupation in the development of AI techniques. There has been a continuum of research based on the logical analysis of text which has resulted in delivered systems which provide creditable performance at 'understanding' the meaning of sentences. But human communication is very complex and even the most powerful NLP systems can still be defeated. In part this is because we humans use both a vast store of knowledge (Popper's World 3) and are grounded in the real physical world. These issues are being addressed head-on by the AI research community.

4.2 Voice recognition/synthesis

Spoken language is the most natural way for humans to communicate. In fact as Wheddon notes [9] speech and language are 'the very essence of social development and have become the key components lying at the humanising heart of the information revolution'. Today speech synthesis is commonplace — automated home banking, information messages from the telephone network, talking computer games to name but a few.

A great deal of work has been carried out to ensure the quality of computer-based text-to-speech systems is acceptable. In fact, powerful commercial systems are becoming available. Laureate [10], developed at BT Laboratories is representative of today's high-quality systems. Speech recognition systems are making equally impressive progress. Single user systems [11] have the capability of recognising many thousands of words following a training session which typically takes an hour or so. Multi-user applications where the system is required to recognise a more limited vocabulary from very many individuals are beginning to make their presence felt in the IT market-place.

What these technological advances demonstrate is that the human user increasingly is being freed from the constraints of textual communications (keyboard and screen) with computers. And this new freedom is providing an interface which is essential for intelligent applications.

4.3 Representation and reasoning

While the graphical and speech capabilities of present-day computer systems have an immediate appeal and are the showcase for modern information technology, it is the inner core of software where the task at hand is conducted. The following account of current practice does not provide information on mainstream AI research and products, such as expert systems, since there are many sources [12, 13] that provide excellent coverage of the field.

From the outset AI research has been concerned with the twin problems of representing knowledge and reasoning over that knowledge. Until comparatively recently there have been two distinct approaches:

- symbolic AI,

- artificial neural networks.

The first approach is motivated by the thesis that it is possible to extract relevant facts and relationships from the real world, label such information with symbols, and use mathematical logic to derive new information from that symbol system. The history of expert systems shows that, for well-defined partitions of the real world (domains in the language of AI), well-engineered solutions to a wide variety of problems can be developed. This particular branch of inquiry has been extended by building precise descriptive (mathematical) models of the domain and using logic to extract causal relationships, new facts, consequences, etc, from the model. What characterises these methods of delivering intelligent software systems is that the designer must extract the relevant facts and relationships from the real-world domain. This so-called knowledge-engineering task is non-trivial for all but toy examples.

The artificial neural network community sets about the task of extracting domain information in a completely different manner. Here the AI engineer is motivated by an understanding of how biological nervous systems operate. One of the prime drivers is that biological systems are not programmed in the same way as computers. In higher order biological systems many capabilities are learned — wired neither at conception nor birth. This, as noted in section 3, provides the plasticity for an organism to adapt to its ever changing environment. Increasing success in this systems engineering approach to intelligent software will free the programmer from having to specify all the subtleties (search space, conditions, data values, etc) of the cognitive engine.

4.4 Soft computing

However, the promise of ANN techniques to deliver more human-like intelligent software has been relatively slow and the application areas patchy. This situation has caused many research groups to investigate alternative routes. Soft computing [14] represents one of the best defined of these alternatives and brings with it a refreshingly ‘engineering’ approach. Soft computing is concerned with the goal of human-oriented computing and not with any particular branch of AI. It will combine techniques as diverse as fuzzy logic, neuro computing, genetic programming, stochastic computing, etc, in a heady, eclectic mix to achieve its goals; and these goals are to make intelligent software systems that are tolerant of imprecision, where the data, and indeed the program, do not have to be 100% accurate or complete. In short, to make cognitive systems which deal with the real world and not some hard, well-bounded and well-specified subset thereof.

5. Pointers to future success

Before computer systems that embody AI techniques are accepted as intelligent — intelligent in the way humans choose to define their cognitive abilities — there are a number of important problems to be solved. These problems can be discovered by analysing how humans grow intellectually. Although we still do not have a detailed understanding of how the human brain operates, we can state with some high degree of certainty that humans develop in the presence of four cognitive influences:

- direct experience of the world (grounding),
- guided learning (parents, teachers),
- exposure to global knowledge (Poppers World 3),
- self-criticism (Poppers World 2?).

The computer equivalent of these human developmental influences are now active research fields in AI.

5.1 Grounded systems — direct experience of the world

Most of the work in artificial intelligence is based on the symbol system hypothesis which posits that a cognitive engine operates on a set of symbols which have a defined relationship with the real world. But as made clear by Brooks [15] such symbols are a pale reflection of reality. For instance, if an expert system were reasoning about the solar system, the symbol ⊙ hardly conveys the size, energy and central importance for life on planet Earth of the Sun! Brooks reasons that in order for systems of artificial intelligence to truly be able to ‘understand’ the real world they must interact, through sensors, with that world and build their knowledge in an incremental fashion. His subsumption architecture provides a framework to develop systems in this bottom-up manner. This approach to building intelligent systems (sometimes called nouvelle AI) has a large measure of scientific and engineering credence. On the one hand, it can be considered to mirror biological evolution — the best existence theorem for intelligent systems; following the Darwinian hypothesis, life on Earth evolved from simple fragments of self-replicating biological material to very complex organisms such as ourselves. On the other hand, subsumption-based systems only have one model of the world — the world itself; and this model is the best there is. While such an approach appeals to engineers such as ourselves, we do not underestimate the problems of building subsumption-based AI systems which will be able to converse with humans in natural speech.

Another approach which tackles the grounding issues is that being pursued by Aleksander. His system, MAGNUS [16], is a finite-state automata approach to neural systems. MAGNUS learns to associate words and images in order to build a comprehensive understanding of its world. Its depth of grounding is not so fundamental as the robots being developed in Brook's Laboratory, but nevertheless is far removed from the symbolic approach of traditional AI. MAGNUS stores its knowledge as a set of states on random access memory — like elements modelled on a general-purpose computer.

5.2 Global knowledge

At the present stage of AI systems development, no single computer program has access to a data store as vast and wide ranging as that possessed by humans. However, research into interface and co-operative agents [17] is one indication of how this situation would change in a relatively short space of time. The World Wide Web provides access to a vast global warehouse. As automatic information extraction and precision techniques gain maturity we could well find intelligent software systems developing their own World 3. And this computer World 3 may not be identical with our own unless we intervene in suitable ways.

One obvious route to ensuring that intelligent software systems do exercise only knowledge with which we are content is to hand-craft the knowledge base.

For the past ten years or so the goal of project CYC has been to encode human common sense knowledge into a computer program. Guha and Lenat [18] describe how this common sense knowledge base can be used to aid information management. Since CYC 'knows' a great deal about our world and relationships between people, it is able to undertake complex data validation tasks without being specifically instructed what to investigate. The authors describe CYC's ability to check databases and spreadsheets for consistency. What this common sense knowledge base confers on the application it supports is a human-like capability to take account of a wide range of anthropomorphic facts and causal relationships.

Lenat [19] notes that CYC has encoded into its knowledge base about 100 000 general concepts spanning human reality and about 1000 000 common sense axioms. This knowledge base is not one vast homogenous structure. Facts and relationships are clustered into contexts and each context has its associated micro-theory. Importantly, CYC's assertions do not have certainty factors, instead CYC assumes each atom of knowledge is true; CYC does not have completeness of inference. What these facts mean is that CYC is a practical demonstration of a system which encodes human common sense knowledge. It uses theoretical approaches to guide construction but does not permit rigour to obstruct progress.

5.3 Computer self-awareness

There is a growing body of opinion [20] that in order for computer programs to be able to display credible intelligence they must have a measure of self-awareness; but currently most computer systems (with the exception of Brook's robots) have hardly any sensory perception at all. They have no plasticity through which they can learn any repertoire of muscular habits. They have not yet been designed to co-ordinate sensory and motor nerves, they cannot hear themselves talk such that they may learn to talk in a particular way. Consider the following situation. In many expert system applications, a computer is programmed to utter statements such as "I did …" or "I said to you …". Such self-referential statements refer to what the program may have done by sending some data to its output buffer, but not what the computer actually did. For human-like self-reference what is needed is for the computer to perceive, see or observe its own action, recognise that the action was indeed its own as distinct from that of any other entity in its environment, and attribute the responsibility for that action on itself. (The 'I' referred to in the above utterances is mostly in the third person.) This is then the beginning of a semblance of self-awareness. An action produced in response to some sensory input is blind until it itself is sensed and owned up to, whether it subsequently results in a self-referential utterance or not. Indeed the subsequent utterance may be different from what was sensed, thus producing a lie.

For computer consciousness much more than self-awareness is needed. For example, an action cannot be seen simply as a response to some sensual stimulus, but as itself being intended to act as a stimulus upon the outside world. This calls for an adequate model of the world in order to decide what sort of stimulus is appropriate at any given stage.

There is thankfully a considerable amount of research in this area, for example, speech-acts-based research is intended to investigate just such behaviour. The point to be made here is that sensing each one of the machine's own actions, reflecting upon it, monitoring its effect, comparing that effect with the intended effect, and so on, are all essential requirements for any intelligent behaviour.

6. Discussion

In section 4 we demonstrated that the current state of art in artificial intelligence and IT is in the process of providing an increasingly human-centred computer environment. We choose the term human-centred because the systems are addressable through speech and respond with speech and high-quality graphical images. Progress in this human/computer interface will be accelerated as computers are welcomed in the home and office alike. We readily admit that high-quality and intuitive interfaces do not of themselves make a computer system intelligent, but they do provide accessibility.

Section 5 provided the reasons why we believe software systems are approaching a level of performance that should be considered intelligent. We set out four influences that help shape human intellectual development and demonstrated how AI research is building machine equivalents. It is, of course, not possible to state that this research is mature — far from it. But progress is incremental and monotonic. There have been few of the setbacks that characterised early AI research. It now appears very unlikely that a new Lighthill [20] will reverse the trend.

7. Conclusions

When viewed from our engineering standpoint the dispassionate observer should be able to recognise the immense strides that are being made in the field. The broad incremental advances in AI now provide an impressive array of techniques to implement intelligent software systems; and the modern, eclectic approach to AI engineering, as evidenced, for example, by the soft computing community, will accelerate the trend to ever-increasing levels of machine intelligence. It falls upon system engineers to harness this new form of intelligence for the benefit of man-

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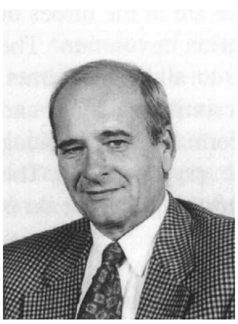

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Professor Abe Mamdani is well known for his research into fuzzy logic which started in the early 1970s. His other research interests include investigations on reasoning under uncertainty and knowledge based systems for industrial control. He has been actively involved with several UK industrial companies in the research and development of applications of knowledge based systems. He has been a long-standing collaborator with the Intelligent Systems Research group at BT Laboratories.

He holds a newly created Chair of Telecommunications Strategy and Services at Imperial College, London. He is a Fellow of the Royal Academy of Engineering and of the Institute of Electrical and Electronic Engineering.